

Presupposition and Priority Modals

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Abstract The Samaritan Paradox (Prior, 1958) is often taken as evidence that deontic *ought* is non-monotonic. In particular, many, following Åqvist (1967), take the Samaritan Paradox to show that $\ulcorner \text{ought } (\varphi \wedge \psi) \urcorner$ does not entail $\ulcorner \text{ought } \varphi \urcorner$. I argue that this conclusion is wrong. The Samaritan Paradox is really about presupposition; in particular, I argue that, when a presupposition trigger occurs in the prejacent of a priority modal (i.e., a preference-related root modal), the priority modal's domain is restricted to worlds satisfying the presupposition. In addition to solving the Samaritan Paradox, the resulting theory entails solutions to many other puzzles concerning priority modals, including Åqvist's Paradox, the Paradox of Gentle Murder, and Chisholm's Paradox.

Keywords: deontic modals; desire ascriptions; presupposition; local contexts

1 Introduction

It's standard to think that $\ulcorner \text{ought } \varphi \urcorner$ means (roughly) that, in all the deontically ideal worlds, φ is true. Following Kratzer (1981, 1991, 2012), we can formally implement this idea in terms of a circumstantial modal base $f(w)$, a set of worlds compatible with the contextually relevant facts, and a pre-order $\succsim_{g(w)}$ on worlds in that modal base generated by an ordering source $g(w)$, a set of propositions representing laws, moral prescriptions, and the like.¹ Given this definition, we can

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¹We can define the pre-order induced by ordering source as follows:

define the domain of *ought* as the set of undominated worlds in the modal base:

$$\text{BEST}(w, f, g) := \{v \in f(w) : \neg \exists u \in f(w)[u \prec_{g(w)} v]\}$$

We can then analyze *ought* as a universal quantifier over this set of undominated worlds:

$$(1) \quad \llbracket \text{ought} \rrbracket^{w, f, g} = \lambda p_{\langle s, t \rangle}. \forall w' \in \text{BEST}(w, f, g) : p(w') = 1$$

Given this semantics, it immediately follows that deontic *ought* is upward monotonic:²

$$(2) \quad \textbf{Upward Monotonicity (UM).} \text{ If } \varphi \models \psi, \text{ then } \bigcirc \varphi \models \bigcirc \psi$$

An UM semantics for deontic modals leads to many well-known puzzles. Prior (1958) observed that, when coupled with a Russellian analysis of definite descriptions, the naïve semantics for *ought* predicts that the inference in (3) is valid.³

$$(3) \quad \begin{array}{ll} \text{a.} & \text{The murder ought to be solved.} & \bigcirc[\exists x[\textbf{murder}(x) \wedge \textbf{solved}(x)]] \\ \text{b.} & \not\models \text{There ought to be a murder.} & \bigcirc[\exists x[\textbf{murder}(x)]] \end{array}$$

This is the Samaritan Paradox – the existence entailment of the definite is not inherited under entailment as predicted by UM. Since Åqvist (1967), the Samaritan Paradox has often been blamed on the fact that a UM semantics validates the following inference: $\bigcirc(\varphi \wedge \psi) \models \bigcirc \varphi$. However, this diagnosis is clearly incorrect – a speaker who utters (3a) does not assert that a person ought to *both* be murdered and helped. Instead, a speaker who utters (3a) *presupposes* that someone was murdered and *asserts* that they ought to be helped.⁴ If we assume a Strawsonian analysis of definites, rather than a Russellian analysis, a natural explanation of (3) emerges: deontic modals only quantify over

$$v \prec_{g(w)} u \Leftrightarrow \{p \in g(w) : u \in p\} \supseteq \{p \in g(w) : v \in p\}$$

While Kratzer does not make this assumption herself, we will make the simplifying assumption that this pre-order is well-founded. That is, for any non-empty modal base $f(w) \neq \emptyset$, $\text{BEST}(w, f, g) \neq \emptyset$. This assumption, equivalent to Lewis’s “Limit Assumption” (1973, §3.4), rules out the possibility of an infinite descending chain of worlds. There is an extensive debate about this assumption, but these questions are orthogonal to the current project (Kaufmann, 2017).

²I will use $\bigcirc$ as a metalanguage operator for *ought* and *should*. The specific instance of UM involving *ought* is sometimes called ‘Inheritance’ in the philosophical literature (Cariani, 2013; Blumberg and Hawthorne, 2021).

³Prior’s original example, as Portner (2009) notes, involves a non-restrictive relative clause in the scope of *ought*. Since non-restrictive relative clauses introduce many syntactic and semantic complexities irrelevant to the current paper, I’ve chosen a simplified variant of Prior’s original example.

⁴This has become a common gloss of the puzzle in linguistics and philosophy of language (Portner, 2009; Cariani, 2013; Booth, 2022) though the point appears underappreciated within deontic logic and philosophy more broadly.

the worlds that satisfy the presuppositions of their prejacent.

This general hypothesis seems well-positioned to explain other paradoxes frequently attributed to UM. For example, Åqvist (1967) observed that when *ought* is coupled with the usual assumption that knowledge is veridical, i.e., the assumption that *know* entails the truth of its complement, an UM semantics for *ought* predicts that, if it ought to be known that φ , then it ought to be that φ . However, as Åqvist's example in (4) illustrates, this entailment is not attested in natural language – the night-guard is obligated to know about a robbery even though the robbery itself is impermissible.

- (4) a. It ought to be that the night-guard knew about the robbery.
 b. $\nRightarrow$ It ought to be that there was a robbery.

While Åqvist viewed his puzzle as a puzzle of entailment, we can also view this puzzle as a puzzle of presupposition. While *know* is veridical, it's also factive – it presupposes the truth of its complement clause.⁵ Two pieces of evidence support my hypothesis that Åqvist's Paradox arises from factivity rather than veridicality. First, we can replicate Åqvist's observation with seemingly all factive attitude verbs, including *regret* and *realize*.

- (5) a. Susie should regret punching Timmie.
 $\nRightarrow$ Susie should have punched Timmie.
 b. Carol should realize that she offended Tom.
 $\nRightarrow$ Carol should have offended Tom.

Moreover, as (6) illustrates, Åqvist's Paradox appears not to arise with attitude verbs like *demonstrate* and *prove*, both commonly thought to be veridical, but non-factive.⁶

⁵While philosophers and logicians often use the terms 'veridical' and 'factive' interchangeably to describe an operator which entails the truth of its prejacent, linguists typically reserve the term 'factive' for operators which *presuppose* the truth of their prejacent (Kiparsky and Kiparsky, 1971; Karttunen, 1971, a.o.).

⁶It's unfortunately very difficult to find attitude verbs which are uncontroversially veridical and non-factive. For example, Degen and Tonhauser (2022) present experimental evidence which is consistent with *demonstrate* being a non-veridical and non-factive.

- (6) a. John should {demonstrate, prove} that he feels apologetic.
 b. $\Rightarrow$ John should feel apologetic.

There are, of course, apparent counterexamples to an UM semantics for deontic modals which appear to have nothing to do with presupposition. As Ross (1944) observed, an UM semantics for deontic modals coupled with Boolean disjunction validates the absurd inference in (7).

- (7) a. John ought to mail the letter.
 b. $\nRightarrow$ John ought to mail the letter or bomb the post office.

Ross’s Paradox and similar cases like Jackson and Pargetter’s (1986) Professor Procrastinate puzzle may ultimately militate against a monotonic theory of deontic modals. However, if my hypothesis is correct, neither the Samaritan Paradox nor Åqvist’s Paradox nor any similar paradoxes give us compelling reason to abandon a monotonic semantics for deontic modals, because such puzzles can ultimately be blamed on presupposition.

Moreover, the general dialectic of this paper naturally generalizes to desire ascriptions. Given the numerous parallels between deontic modals and bouletic attitude verbs, I will, following Portner’s (2009) lead, lump them together under the single header of “priority modals.” If we give bouletic attitude verbs a Hintikka-style (1969) relational semantics, then $\ulcorner S \text{ wants } \varphi \urcorner$ is true just in case φ is true in every world compatible with S ’s desires.

$$(8) \quad \llbracket \text{want} \rrbracket^w = \lambda x_e. \lambda p_{(s,t)}. \forall w' \in \text{DES}(x, w) : p(w') = 1$$

where $\text{DES}(x, w) = \{w' : \text{all of } x\text{'s desires in } w \text{ are satisfied in } w'\}$

Since the naïve Hintikkan semantics also validates UM, if we assume a Russellian analysis of definite descriptions, it too generates bouletic analogues of the Samaritan Paradox.

- (9) a. John wants the murder to be solved. $\text{WANTS}_j[\exists x[\mathbf{murder}(x) \wedge \mathbf{solved}(x)]]$
 b. $\nRightarrow$ John wants there to be a murder. $\text{WANTS}_j[\exists x[\mathbf{murder}(x)]]$

Like its deontic counterpart, bouletic versions of the Samaritan Paradox have been blamed on the fact that a UM semantics for *want* validates a conjunctive inheritance entailment: $\text{WANTS}(\varphi \wedge \psi) \models \text{WANTS}(\varphi)$. Here too this diagnosis seems clearly incorrect – a speaker who utters (9a) clearly does not assert that John wants someone to *both* be murdered, and their murder solved. Instead, a speaker

who utters (9a) *presupposes* that someone was murdered and *asserts* that they ought to be helped. Here too if we assume a Strawsonian analysis of definites we can predict the failure of UM given the assumption that *want* only quantifies over worlds where the presuppositions of its complement are satisfied.⁷

The most famous counterexamples to a monotonic analysis of bouletic attitude verbs involve complements with modifiers, such as the adverb *peacefully* in Stalnaker’s example (10).

- (10) a. Alice wants to die peacefully. (Stalnaker, 1984)
 b. $\nRightarrow$ Alice wants to die.

While these failures of UM may seem difficult to attribute to presupposition, Crnič (2011) observes that relevant modifiers do license inferences which project similarly to presuppositions:

- (11) a. Alice didn’t die peacefully. $\rightsquigarrow$ *Alice died* [negation]
 b. Did Alice die peacefully? $\rightsquigarrow$ *Alice died* [question]
 c. Alice might have died peacefully. $\rightsquigarrow$ *Alice died* [modal]

Following Geurts and Sandt (2004), Abrusán (2013), and others, I will assume that modifiers like *peacefully* are focus-marked by default. According to many standard theories of focus, focus-marking introduces an existential presupposition – namely, focus seems to presuppose that the sentence is true given some alternative to the focus-marked constituent.⁸ For example, while (12) has the same truth-conditions regardless of whether *John* or *pasta* is focus-marked, what (12) presupposes clearly depends on which constituent we decide to focus-mark:

- (12) John ate pasta.
 a. John ate PASTA. $\rightsquigarrow$ *John ate something*
 b. JOHN ate pasta. $\rightsquigarrow$ *someone ate pasta*

Hence, if we assume that there’s inherent focus-marking on *peacefully* and focus-marking triggers

⁷Indeed, the fact that definites embedded under desire ascriptions do not seem to entail an existential desire has been used by many authors to argue against a Russellian analysis (Heim, 1991; Kripke, 2005; Elbourne, 2005, 2010).

⁸While a purely presuppositional analysis of focus, first proposed by Chomsky (1970) and more recently revived by Geurts and Sandt (2004), has largely fallen out of favor due to Jackendoff’s (1972) forceful criticisms, the influential Hamblin-style (1976) alternative semantics for focus proposed by Rooth (1985, 1992) also predicts that focus-marking carries an existential presupposition. In this paper, I won’t attempt to adjudicate between these analyses; instead, I simply rely on their shared prediction that focus-marking carries an existential presupposition.

an existential presupposition, then (10) presupposes that there's some way in which Alice will die, and asserts that Alice wants that way in which she dies to be peaceful.

In this paper, I take this basic observation that failures of UM often arise due to presuppositions, to its logical conclusion. In §2, I introduce the satisfaction theory – the dominant approach to presupposition in semantics and pragmatics – highlighting several predictions that Schlenker's (2009) version of the satisfaction theory makes concerning the presuppositions of priority modals. Given the results in §2, I argue, in §3, that a non-monotonic semantics isn't sufficient to solve the Samaritan Paradox, and that the popular DIVERSITY-based solution rests both on ad hoc assumptions and has clear empirical shortcomings. Finally, in §4, I develop a positive solution to the Samaritan Paradox based on the idea that priority modals quantify over a “local context” which satisfies their prejacent's presuppositions. The resulting theory entails solutions to a host of well-known deontic paradoxes, including Åqvist's Paradox, the Paradox of Gentle Murder, and Chisholm's Paradox. It also has several other desirable properties – it predicts failures of disjunctive syllogism involving priority modals and it entails that priority modals are focus-sensitive. However, I also highlight a drawback of my approach: it leads to the proliferation of Zvolenszky's (2002) problem – in addition to replicating the restrictor analysis's prediction that “if φ , ought φ ” is trivially true, the resulting account also predicts that “ φ and ought φ ” and “either $\neg\varphi$ or ought φ ” are trivially true.

2 A crash course on presupposition

The textbook theory of presupposition projection is the *satisfaction theory* developed in the 1970s by Stalnaker (1974, 1978) and Karttunen (1973, 1974).⁹ The satisfaction theory starts by assuming that presuppositions are preconditions on the *context set*, the set of worlds compatible with the propositions that are common ground. For example, a speaker can felicitously utter a factive attitude report like *S* knows that φ only if φ holds throughout the context set – i.e., if $c \subseteq \llbracket\varphi\rrbracket$. However, Karttunen and Stalnaker realized that, to adequately capture “presupposition filtering” in complex sentences, the global context often needed to shift mid-sentence. For example, consider

⁹For reasons of space, I won't discuss trivalent theories (Peters, 1979; George, 2008a,b) or multidimensional theories (Karttunen and Peters, 1979; Sudo et al., 2012) which are also popular in contemporary semantic theory.

the following minimal pair:

- (13) a. If John once was a smoker, then he stopped smoking.
 b. If John contracted emphysema, then he stopped smoking.

While the consequent’s presupposition that John once smoked projects in (13b), it doesn’t project in (13a); hence, the sentence as a whole presupposes nothing. We can explain this contrast by allowing the global context to shift when we evaluate the conditionals in (13). Hence, we evaluate the consequent of both conditionals in a “local context” obtained by updating the global context set with the information in the conditional’s antecedent.

While neither Karttunen nor Stalnaker developed a rigorous, compositional account of local contexts, Heim (1983) would provide such an account couched in dynamic semantics. In particular, we can obtain the desired local contexts by recursively defining the following update rule $\cdot[\cdot] : \mathcal{C} \times \mathcal{L} \rightarrow \mathcal{C}$, which maps a context and a sentence to a new context:

$$\begin{aligned} c[p] &:= \{w \in c : w(p) = 1\} \\ c[\neg\varphi] &:= c \setminus c[\varphi] \\ c[\varphi \wedge \psi] &:= c[\varphi][\psi] \\ c[\varphi \vee \psi] &:= c[\varphi] \cup c[\neg\varphi][\psi] \end{aligned}$$

While Heim’s dynamic implementation makes a host of desirable predictions, Soames (1989) argued that theory wasn’t explanatory. In particular, since the update rules for the connectives are stipulative, nothing in Heim’s system rules out the possibility of an entry for conjunction where the context set c is first updated with the right conjunct – i.e., $c[\varphi \wedge \psi] = c[\psi][\varphi]$.¹⁰ However, such an update rule would obviously license unattested patterns of projection and filtering.

Schlenker (2009, 2010a,b) attempts to solve the explanatory problem by introducing an algorithm for computing local contexts from an expression’s Logical Form (LF). Schlenker calls a proposition *transparent* in a syntactic environment if including the proposition in that environment does not

¹⁰See Rothschild (2011) for an attempt to develop an explanatory dynamic semantics that avoids this critique.

change the utterance’s overall truth-value no matter how the utterance is continued. For example, consider the following partially finished utterance:

(14) Either John never smoked or _____

Schlenker’s account is premised on the idea that, at this stage, the hearer reasons about what information she can take for granted given the utterance’s possible continuations – i.e., what worlds in the context set are compatible with every way of filling in the blank in (14)? If the first disjunct is false, the hearer reasons that she can add its negation to the second disjunct without altering (14)’s truth-value. Likewise, if the first disjunct is true, the hearer reasons that she can add its negation to the second disjunct without altering (14)’s truth-value because, for any continuation whatsoever, (14) will be true. Hence, we can safely include the proposition John previously smoked (i.e., the first disjunct’s negation) in the second disjunct without altering the sentence’s overall truth-value; hence, the proposition that John previously smoked is transparent in (14)’s second disjunct.

More concretely, Schlenker defines a syntactic environment’s local context as the smallest (i.e., the most informative) proposition that is transparent in the syntactic environment.

Definition 2.1. (asymmetric) local contexts

The *local context* of b occurring in a syntactic environment a_c as the smallest proposition p such that $\llbracket a(p \wedge b')c' \rrbracket \Leftrightarrow \llbracket ab'c' \rrbracket$ for all completions b' and c' of the appropriate type.

Given an arbitrary local context κ , Schlenker’s algorithm yields the following predictions:

$$\begin{aligned} \llbracket \neg\varphi \rrbracket^\kappa \neq \# &\Leftrightarrow \llbracket \varphi \rrbracket^\kappa \neq \# \\ \llbracket \varphi \wedge \psi \rrbracket^\kappa \neq \# &\Leftrightarrow \llbracket \varphi \rrbracket^\kappa \neq \# \wedge \llbracket \psi \rrbracket^{\kappa+\varphi} \neq \# \\ \llbracket \varphi \vee \psi \rrbracket^\kappa \neq \# &\Leftrightarrow \llbracket \varphi \rrbracket^\kappa \neq \# \wedge \llbracket \psi \rrbracket^{\kappa+\neg\varphi} \neq \# \end{aligned}$$

where $\kappa + \varphi := \{w \in \kappa : \llbracket \varphi \rrbracket^w = 1\}$ and the $\#$ denotes that the interpretation function is undefined.¹¹ Hence, Schlenker’s algorithm replicates the predictions of Heim’s account without

¹¹In other words, I treat $\llbracket \cdot \rrbracket$ as a total function mapping a sentence, local context pairs onto $\{1, 0, \#\}$ rather than treating $\llbracket \cdot \rrbracket$ as a partial function. This choice is purely for convenience.

raising the same explanatory problem.

Given Schlenker’s theory, we can derive concrete predictions concerning the local context of deontic modals. For example, *ought*’s local context κ is the smallest proposition such that for any φ : $\bigcirc(\kappa \wedge \varphi) \Leftrightarrow \bigcirc\varphi$. If we assume the naïve Kratzerian semantics in (1), this yields the following prediction about *ought*’s local context:

$$(15) \quad \kappa = \bigcup \{ \text{BEST}(w, f, g) : w \in c \}.$$

This prediction alone has problematic consequences. For example, while a negated factive attitude ascription does not *entail* its complement, it still *presupposes* its complement.¹² This alone suffices to generate instances of Åqvist’s Paradox. Consider the following non-entailment:

- (16) a. John should not know that he’s getting a surprise.
 b. $\nRightarrow$ John should get a surprise.

The negated factive attitude verb in (16a) presupposes that John is getting a surprise; hence, Schlenker’s algorithm coupled with the naïve Kratzerian semantics predicts that all the deontically ideal worlds accessible from the context set are worlds where John gets a surprise. Hence, the naïve Kratzerian semantics when combined with a fairly orthodox theory of presupposition projection predicts that (16a) entails (16b).

Schlenker’s theory also yields concrete predictions concerning the local context of bouletic attitude verbs – an issue that has been explored in detail by Blumberg and Goldstein (2023). If we assume a naïve Hintikkan semantics for *want*, then the local context $\kappa = \bigcup \{ \text{DES}(S, w) : w \in c \}$ – i.e., the local context of *want* is its entire modal domain. This prediction alone also suffices to generate many Samaritan-like inferences. Consider the following non-entailment:

- (17) a. Mary wants to get well.
 b. $\nRightarrow$ Mary wants to be sick.

Given Schlenker’s algorithm coupled with a naïve Hintikkan semantics, (17a) presupposes that all the worlds compatible with Mary’s desires accessible from c are worlds where Mary is sick; hence, we incorrectly predict that (17a) should entail (17b).

¹²I owe this argument to REDACTED (p.c.)

A similar problem arises for other semantics for *want*. For example, von Fintel (1999) adopts a Kratzer-style semantics for *want* where it's evaluated with respect to a doxastic modal base f_{DOX} and a preference ordering $\succsim_{g(w)}$ on worlds.¹³

$$(18) \quad \llbracket \text{want} \rrbracket^{w,f,g} = \lambda x_e. \lambda p_{\langle s,t \rangle}. \forall w' \in \text{BEST}(x, w, f_{\text{DOX}}, g) : p(w') = 1$$

Given von Fintel's semantics for *want*, the complement's local context $\kappa = \bigcup \{ \text{BEST}(S, w, f_{\text{DOX}}, g) : w \in c \}$ is again the entire modal domain. Hence, the invalid inference in (17) is also validated by von Fintel's semantics.

3 Against extant solutions

There are two main approaches to explaining Samaritan Paradox-like failures of UM. The first approach gives priority modals a non-monotonic semantics. As I show in this section, given plausible assumptions about presupposition, this move alone fails to block many instances of the Samaritan Paradox. The second approach posits that priority modals presuppose that their modal base is *diverse* – i.e., contains worlds where the prejacent is true, and worlds where the prejacent is false. I highlight several problems with this strategy – in particular, it rests on ad hoc assumptions concerning the local contexts of a priority modal's prejacent and it incorrectly predicts that (counter)factual priority modals are always infelicitous.

¹³The choice of a doxastic modal base is motivated by the fact that desire ascriptions give rise to *belief projection*. When an attitude verb contains a presupposition trigger in its complement, like (i), it often generates two inferences: an inference that the attitude holder believes the presupposition, and the inference that the presupposition is actually true. Following Geurts (1998), I call these two inferences the *i-inference* and *e-inference*, respectively.

- | | | |
|---|---|-------------|
| i | Ann wants Bill to stop smoking. | |
| | $\rightsquigarrow$ Anne believes Bill smokes. | i-inference |
| | $\rightsquigarrow$ Bill smokes. | e-inference |

von Fintel's semantics does not predict the i-inference. Rather, it predicts something weaker; for example, in (i), it predicts that Anne believes Bill possibly smokes. It's hard to encode the i-inference in the truth-conditions without breaking things elsewhere. This is a specific instance of the more general *proviso problem* which plagues the satisfaction theory (Geurts, 1996) – see Sudo (2014) for further discussion of the problem.

3.1 Strategy #1: go non-monotonic

Given minimal assumptions about how presuppositions interact with priority modals, it's clear that a non-monotonic semantics is insufficient to solve the Samaritan Paradox. While a non-monotonic semantics ensures that priority modals aren't closed under entailment, it's still possible that they might be closed under presupposition. Consider Heim's (1992) non-monotonic semantics for *want* inspired by Stalnaker's (1984) similar proposal:

$$(19) \quad \llbracket S \text{ wants } \varphi \rrbracket^w = \forall w' \in \text{DOX}_{S,w} : \mathbf{sim}(w', \llbracket \varphi \rrbracket \cap \text{DOX}_{S,w}) > \mathbf{sim}(w', \llbracket \neg \varphi \rrbracket \cap \text{DOX}_{S,w})$$

where $\mathbf{sim}$ is a Stalnaker-style (1968) selection function and $>$ is a preference ordering on worlds. In words, Heim's semantics yields the following truth-conditions: ' S wants φ ' is true just in case the closest φ -world compatible with the attitude-holder's beliefs is better than the closest $\neg\varphi$ -world compatible with the attitude-holder's beliefs.

Heim's semantics, coupled with Schlenker's algorithm, predicts that the local context of *want*'s complement is the worlds compatible with the attitude-holder's beliefs.

$$(20) \quad \kappa = \bigcup \{ \text{DOX}_{S,w} : w \in c \}$$

If we assume that $\mathbf{sim}(w, \emptyset) = \emptyset$, i.e., that contradictions are not true in any closest world, then Heim's semantics incorrectly predicts that the inference in (17) is trivially true. To avoid this problem, Heim stipulates $\mathbf{sim}(w, \emptyset)$ is undefined – in essence, this amounts to a version of the DIVERSITY requirement that I will criticize in the next section. However, absent this stipulation, Heim's non-monotonic semantics fails to explain many instances of the Samaritan Paradox.

Jackson's (1985) semantics for deontic modals, which has recently received renewed attention among semanticists due to Chung's (2019, 2020) work, mirrors Heim's semantics for *want*:

$$(21) \quad \llbracket \text{ought } \varphi \rrbracket^w = \forall \psi \in \text{Alt}(\varphi) : \mathbf{sim}(w, \llbracket \varphi \rrbracket \cap f(w)) > \mathbf{sim}(w, \llbracket \psi \rrbracket \cap f(w))$$

where $\text{Alt}(\varphi)$ denotes the relevant alternatives to φ . Jackson's semantics predicts that ' $\text{ought } \varphi$ ' is true just in case the way φ would happen given the circumstances is preferable to the way in which any alternative to φ would happen given the circumstances. When combined with Schlenker's algorithm, the local context of *ought*'s prejacent is $\kappa = \bigcup \{ f(w) : w \in c \}$ – i.e., the circumstantially

possible worlds compatible with the context set. Hence, if we assume that $\mathbf{sim}(w, \emptyset) = \emptyset$, i.e., that contradictions are not true in any closest world, then Jackson’s semantics incorrectly predicts that the inference in (16) is trivially true. Therefore, without additional assumptions, a non-monotonic semantics isn’t sufficient to solve the Samaritan Paradox.

3.2 Strategy #2: Diversity

The textbook solution to the Samaritan Paradox – proposed by Heim (1992) and von Fintel (1999) and elevated to textbook status in von Fintel and Heim (2011) – combines two assumptions.¹⁴ First, the received solution assumes that the modal base is the local context of *ought*’s prejacent – that is, the modal base must satisfy the presupposition of the modal’s prejacent.

- (22) **Modal Base (MB) Satisfaction.** An assertion of $\ulcorner \text{ought } \varphi \urcorner$ in context c is felicitous only if the modal base $f(w)$ satisfies the prejacent’s φ ’s presuppositions for all $w \in c$.

Second, the received solution assumes that deontics presuppose that their modal base is “diverse” – that is, it contains worlds where the prejacent is true and worlds where the prejacent is false.¹⁵

- (23) **Diversity.** An assertion of $\ulcorner \text{ought } \varphi \urcorner$ in a context c presupposes that $f(w) \cap \llbracket \varphi \rrbracket \neq \emptyset$ and $f(w) \cap \llbracket \neg \varphi \rrbracket \neq \emptyset$ for all $w \in c$.

Given DIVERSITY and MB SATISFACTION, we can then derive the failure of the Samaritan Inference. In (3a), the definite in *ought*’s prejacent presupposes that someone was murdered. Given PREJACENT SATISFACTION, this presupposition must hold throughout the modal base, but this means that (3b) will violate DIVERSITY. In the bouletic context, von Fintel (1999) adopts a parallel strategy. von Fintel’s semantics predicts that *want* is UM; however, he assumes *want* is subject to both MB SATISFACTION and DIVERSITY which allows him to give an analogous explanation of the Samaritan Paradox for bouletic attitude verbs. I believe there are many problems with this strategy.

First, the MB SATISFACTION assumption doesn’t fall out of a predictive theory of presupposition

¹⁴What I’m calling the “received” solution is rarely spelled out in detail by its proponents, the two clearest statements of the view can be found in Carr (2014) and Booth (2022) who are both critical of the view.

¹⁵This was dubbed the “diversity condition” by Condoravdi (2002) who originally posited the principle to explain differences in temporal orientation between epistemic and root modals.

projection. Indeed, as we saw in §2, Schlenker’s algorithm predicts that the local context of a priority modal is its *modal domain* rather than its *modal base*. However, we cannot reformulate DIVERSITY as a requirement imposed on modal domains without making all strong priority modals trivially false. Hence, a DIVERSITY-based solution to the Samaritan Paradox seemingly rests on an assumption which is extremely hard to justify on independent grounds.

However, there’s a far more basic, empirical problem with this strategy. If priority modals were subject to DIVERSITY, then they would have neither factual nor counterfactual uses of priority modals ought to be felicitous. Heim (1992) herself flags this issue, conceding that “it doesn’t seem right that one can never speak of wanting things one is convinced will happen or convinced won’t happen” (p. 199). In natural language, it’s perfectly felicitous to use deontic modals and desire ascriptions counterfactually – i.e., in a context where you know the argument of the priority modal is not true in any accessible world. It’s an unfortunate fact of life that the deontically ideal state-of-affairs is often circumstantially impossible:

- (24) **Context:** The speaker must choose between diverting an out-of-control trolley, killing a single bystander, and letting five people die.
- a. Everyone ought to be saved, but, given the circumstances, it’s not possible.
 - b. It’s impossible to save everyone, but everyone ought to be saved.

Likewise, it’s another unfortunate fact of life that many of our desires are impossible:

- (25) a. Even though it’s impossible, I want this weekend to last forever.
 b. I want 91 to be a prime number, even though I know it’s composite.

Conversely, it’s perfectly felicitous to use deontic modals and desire ascriptions factually – i.e., in a context where you know the argument of the priority modal is true in all accessible worlds.

- (26) a. John paid his taxes because John knows he ought to pay his taxes.
 b. I live in Bolivia because I want to live in Bolivia. Iatridou (2000)

In the deontic case, there’s a subtlety with the data – whether an *ought* can be used counterfactually depends on whether it’s evaluative or deliberative. An evaluative *ought* describes the way things

ought-to-be, while a deliberative *ought* describes what an agent ought-to-do.¹⁶

- (27) a. There ought to be world peace. [evaluative]
 b. John ought to bring about world peace. [deliberative]

Unlike evaluative *ought*, deliberative *ought* is usually taken to imply *can*; hence, deliberative *ought* is infelicitous when used counterfactually. While one might attempt to restrict the scope of DIVERSITY to deliberative *oughts*, I believe that this move will ultimately fail for two reasons.¹⁷ First, the Samaritan Paradox arises for both evaluative *oughts* too, e.g., it's completely coherent for a speaker to accept (28a) but reject (28b), and this would need to be explained independently.

- (28) a. There ought to be a stop to the war with Iran.
 b. $\nrightarrow$ There ought to be (an ongoing) war with Iran.

Second, it seems perfectly coherent to use a deliberative *ought* even in a context where the prejacent is true in every circumstantially possible, as examples like (26a) illustrate.

While I can't conclusively prove that there's no revised version of DIVERSITY that avoids the problem posed by (counter)factual priority modals, I'm skeptical about the viability of such views. In the context of desire ascriptions, Rubinstein (2017) proposes a modification of DIVERSITY where it's relativized to a contextually circumscribed set of worlds D – i.e., a priority modal with prejacent φ presupposes $D \cap f(w) \cap \llbracket \varphi \rrbracket \neq \emptyset$ and $D \cap f(w) \cap \llbracket \neg \varphi \rrbracket \neq \emptyset$ for all $w \in c$. While this doesn't solve the problem posed by (counter)factual uses, Grano and Phillips-Brown (2022) note that Rubinstein's version of DIVERSITY can capture (counter)factual priority modals if we weaken it and only require that D itself be diverse with respect to φ ; however, as they note, the resulting constraint seems too unconstrained to really be predictive. Grano and Phillips-Brown themselves redefine DIVERSITY in terms of conditional beliefs, treating the difference between (counter)factual desire ascriptions and non-(counter)factual desire ascriptions as akin to the difference between subjunctive and indicative conditionals. However, treating the difference between (counter)factual and non-(counter)factual

¹⁶There's an extensive debate about how this difference between these two readings should be captured. Schroeder (2011), for example, argues that the operative difference between evaluative and deliberative *oughts* is syntactic – whereas deliberative *oughts* are control predicates, evaluative *oughts* are raising predicates. However, see Bhatt (1998) for counter-arguments that deliberative *oughts* are raising.

¹⁷Dowell (2013) seems to suggest a view like this, arguing that, in deliberative contexts, both a deontic modal's prejacent and the negation of its prejacent must be live options.

priority modals as akin to the difference between subjunctive and indicative conditionals strikes me as questionable in light of cross-linguistic data. It’s well-established that many languages use the same morphology – what von Stechow and Iatridou (2023) call X-marking – to mark both subjunctive conditionals and priority modals. For example, in Hungarian, the same morpheme *-nA* marks subjunctive conditionals, bouletic attitude verbs, and deontic necessity modals.

The failure of Diversity-based accounts suggests that the crucial ingredient is not diversity in the modal base, but rather the interaction between modal evaluation and presupposition satisfaction. The next section develops this idea formally.

4 An indirect restriction theory of priority modals

My positive analysis starts with a fairly uncontroversial fact: when we embed a trigger in a deontic modal’s prejacent or the complement of a bouletic attitude verb, the associated presupposition always projects – i.e., priority modals are *holes* in Karttunen’s (1974) sense.

- (29) a. John {wants, ought} to stop smoking.
 b. $\rightsquigarrow$ John’s currently a smoker.

In this section, I show that baking this very simple assumption into the semantics of priority modals yields an immediate solution to the Samaritan Paradox and several related puzzles. I will specifically bake the hole-like quality of priority modals into their semantics by allowing any precondition its prejacent imposes on the local context to be “passed up” to the modal.

In particular, when a presupposition trigger occurs in a priority modal’s prejacent, I propose that the priority modal’s modal base is indirectly restricted to a local context satisfying the prejacent’s presuppositions, effectively allowing preconditions imposed on the context by the prejacent to percolate upward, influencing the modal’s interpretation.¹⁸ In other words, for any priority modal $\mathcal{O}$, an assertion of $\ulcorner \mathcal{O} \varphi \urcorner$ is evaluated relative to a modal base f restricted by a local context satisfying

¹⁸Blumberg and Holguín (2019) develop a similar idea to explain readings of attitude reports embedded in conditionals and disjunctions. However, the current project differs in a few respects; in particular, they neither attempt to integrate their account into an explanatory theory of presupposition projection nor do they attempt to explain the deontic paradoxes we’re interested in here.

φ 's presuppositions. The following toy semantics illustrates the proposal:

$$(30) \quad \llbracket \text{ought} \rrbracket^{w,f,g,\kappa} = \lambda p_{\langle s,t \rangle}. \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : p(w') = 1$$

where $f^{+\kappa}(w) = f(w) \cap \kappa$. In words, given a sentence of the form $\lceil \text{ought } \varphi \rceil$, if φ presupposes p , then κ is a local context such that $\kappa \subseteq p$. That is, the domain of *ought* is restricted to a local context which entails the prejacent's presuppositions. In contrast, if φ presupposes nothing, then the local context is simply the trivial local context, i.e., $\kappa \equiv \top$. Since a modal with a prejacent that presupposes nothing will quantify over a larger domain than a modal with a prejacent that presupposes something, the Samaritan Inference will not, in general, be valid. For example, while (3a) quantifies over the best worlds where someone has been murdered, (3b) quantifies over *all* the best worlds; hence, (3a) can be true even when (3b) is false.

von Fintel's Kratzerian semantics for *want* can be similarly modified by requiring that the modal base be directly restricted by a local context satisfying the complement's presuppositions:

$$(31) \quad \llbracket \text{want} \rrbracket^{w,f,g,\kappa} = \lambda x_e. \lambda p_{\langle s,t \rangle}. \forall w' \in \text{BEST}(x, w, f_{\text{DOX}}^{+\kappa}, g) : p(w') = 1$$

This yields a parallel explanation of the bouletic Samaritan Paradox. Since *want* quantifies over the most desirable doxastically accessible worlds compatible with its complement's presuppositions, *want* will quantify over a larger domain when its complement is presupposition-free.

My indirect restriction solution to the Samaritan Paradox has clear antecedents in the deontic logic literature. For example, Tomberlin (1981) thought that (3a) is a covert conditional which he translated using a conditional deontic operator $\bigcirc(\cdot \mid \cdot)$ as (32).

$$(32) \quad \bigcirc(\text{the murder is solved} \mid \text{there is a murder})$$

Like Tomberlin, I assume that (3a) has an implicit conditional meaning – e.g., the truth-conditions I assign *ought* in (30) are precisely the same truth-conditions Kratzer's (1986, 2012) restrictor analysis assigns (32). However, despite this similarity, the solutions have a major difference. Tomberlin's solution leaves the Samaritan Paradox a mystery – it's unclear why (3a), a sentence without any overt conditional syntax or morphology, is interpreted conditionally.¹⁹ In contrast, my account pro-

¹⁹Many authors have noted that theories which treat conditional *oughts* as involving a special deontic conditional operator $\bigcirc(\cdot \mid \cdot)$ are linguistically implausible. Since they do not attempt to derive the meaning of deontic conditionals

vides a linguistically plausible mechanism for deriving this conditional interpretation.

For simplicity, we will treat the local context as a parameter of the index of evaluation. For logical connectives, we will assume the local contexts predicted by Schlenker’s algorithm. For lexical presupposition triggers, such as the change-of-state verb *stop* or the factive attitude verb *know*, I will simply hardwire the precondition they impose on the local context into their denotation:

$$\begin{aligned} \llbracket \text{stop} \rrbracket^{w,f,g,\kappa} &:= \lambda p_{\langle s,t \rangle} \cdot \begin{cases} \text{defined iff } \kappa \subseteq \text{PAST}(p)(w) \\ 1 \text{ iff defined and } \text{stop}(p)(w) = 1 \end{cases} \\ \llbracket \text{know} \rrbracket^{w,f,g,\kappa} &:= \lambda p_{\langle s,t \rangle} \cdot \begin{cases} \text{defined if } \kappa \subseteq p \\ 1 \text{ iff defined and } \text{know}(p) = 1 \end{cases} \end{aligned}$$

In words: the change-of-state verb *stop* checks whether its local context entails that its argument was true in the past, while the factive attitude verb *know* checks whether its local context entails its complement. Now we can derive solutions to the Samaritan Paradox and Åqvist’s Paradox.

First, let’s see how the indirect restriction semantics solves the Samaritan Paradox by working through a concrete example. If we combine my entry for *stop* with my new entry for *ought*, it follows that (33a) has the truth-conditions in (33b). So, (33a) is true just in case, in the deontically ideal worlds where John smoked in the past, John stops smoking.

- (33) a. John ought to stop smoking.
- b. $\llbracket (33a) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \text{PAST}(\text{smoking}(j)) \\ 1 \text{ iff } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \neg(\text{smoking}(j))(w') = 1 \end{cases}$

In contrast, (34a) has the truth-conditions in (34b).

- (34) a. John ought to have smoked.
- b. $\llbracket (34a) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \top \\ 1 \text{ iff } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \text{PAST}((\text{smoking}(j))(w')) = 1 \end{cases}$

from the interaction of *ought* and *if*, we’re essentially treating conditional *oughts* as idioms (Thomason, 1981; Kolodny and MacFarlane, 2010, a.o.)

Since (34a) quantifies over a larger domain than (33a), it's clear that (34a) can be false even when (33a) is true. Moreover, given the indirect restriction semantics, we can offer a similar solution to Åqvist's Paradox. If we combine this analysis of factive attitude verbs with my entry for *ought*, we obtain the following truth-conditions for (4a):

$$(35) \quad \llbracket (4a) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \llbracket \text{there was a robbery} \rrbracket \\ 1 \text{ iff defined and } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \mathbf{know}(j)(w') = 1 \end{cases}$$

However, (4b), in contrast, is restricted by a trivial local context.

$$(36) \quad \llbracket (4b) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \top \\ 1 \text{ iff defined and } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \mathbf{was-a-robbery}(w') = 1 \end{cases}$$

Since (4a) quantifies over a smaller domain than (4b), (4a) does not entail (4b).

While my indirect restriction semantics invalidates UM, there's a sense in which the semantics is still fundamentally monotonic, because monotonicity can only fail in my semantics for presupposition-related reasons; hence, once we factor out the contribution of presupposition, the semantics still validates UM. More precisely, if we let $\pi(\varphi)$ denote what φ 's presupposes, then we can formulate a weakening of UM which our semantics validates:

$$(37) \quad \mathbf{Restricted\ UM.} \text{ If } \varphi \models \psi \text{ and } \pi(\varphi) = \pi(\psi), \text{ then } \bigcirc\varphi \models \bigcirc\psi$$

In words: if φ entails ψ and φ and ψ have the same presuppositions, then an obligation to φ entails an obligation to ψ . For this reason, my indirect restriction semantics constitutes a very conservative extension of the naïve Kratzerian semantics – the two theories yield all the same predictions concerning presupposition-free sentences.

In the remainder of this section, I discuss some desirable consequences of the view presented here. In particular, it derives desirable predictions concerning the interaction of priority modals with disjunctions and conditionals, it straightforwardly predicts that priority modals are focus-sensitive, and it predicts the absence of a Samaritan Paradox for epistemic modals. I conclude by discussing a drawback of the current approach – it implies the existence of a version of Zvolenszky's (2002) problem for disjunction and conjunction.

Application #1: failures of disjunctive syllogism INDIRECT RESTRICTION predicts that right-embedded deontic modals and bouletic attitude verbs in disjunctions should license restricted readings where the modal domain is restricted by the negation of the left disjunct. In this section, I show that this prediction immediately explains why disjunctive syllogism sometimes fails for disjunctions containing priority modals.

First, deontic modals embedded in right disjuncts often appear to quantify over a domain restricted by their left disjunct's negation.²⁰ For example, consider Kolodny and MacFarlane's (2010) miners scenario. Ten miners are trapped either in shaft A or in shaft B, but we do not know which. The two shafts are filling with water. We have enough sandbags to block one shaft, but not both. If we block one shaft, all the water will go into the other shaft, killing any miners inside it. If we block neither shaft, both shafts will fill halfway with water, and just one miner will be killed. In this context, the following instance of disjunctive syllogism fails:

- (38) a. Either the miners are in shaft A, or we should block shaft B. $in_A \vee \bigcirc block_B$
 b. We shouldn't block shaft B. $\neg \bigcirc block_B$
 c. $\nRightarrow$ The miners are in shaft A. in_A

My theory predicts the existence of failures like these. My theory predicts that *should* in (38a) quantifies over a modal base restricted to worlds where the miners aren't in shaft A; however, *should* in (38b) is predicted to quantify over the entire modal base.

von Fintel and Pasternak (2022), building on earlier work by Blumberg and Holguín (2019), present similar data suggesting that bouletic attitude verbs embedded in right disjuncts often appear to quantify over a domain restricted by their left disjunct's negation. They imagine a context where Mary believes that she may have been infected with a virus that will turn her into a zombie. As they observe, the following instance of disjunctive syllogism seems intuitively invalid:

²⁰Dorr and Hawthorne (2013) discuss very similar data concerning failures of disjunctive syllogism in disjunctions containing epistemic *must*. Mandelkern (2024) shows how this can be derived by assuming that the domains of epistemic modals are bounded by their local context.

- (39) a. Either this test is a false positive, or Mary wants to be killed.
 b. The test isn't a false positive.
 c. $\nRightarrow$ Mary wants to be killed.

My theory also predicts that bouletic attitude verbs in disjunctions can lead to failures of disjunctive syllogism. In (39a), *want* quantifies over a modal base restricted to worlds where the test isn't a false positive, while *want* in (39c) quantifies over the entire modal base.

Application #2: focus-sensitivity and the Paradox of Gentle Murder A shared feature of both deontic modals and bouletic attitude verbs is their focus-sensitivity (Villalta, 2008; Finlay and Snedegar, 2014; Harner, 2016). For example, Villalta highlights the contrast in (40).

(40) **Context.** John doesn't want to teach logic this term, but, given that he has to teach logic this term, he would rather teach on Thursdays than on Wednesdays.

- a. John wants to teach LOGIC on Thursdays. [false]
 b. John wants to teach logic ON THURSDAYS. [true]

(40a) is intuitively false because John does not want to teach logic; however, (40b) is intuitively true because, if John has to teach logic, he wants to teach logic on Thursdays as opposed to Wednesdays.

First, the focus-sensitivity of priority modals follows immediately from our indirect restriction semantics given the assumption focus-marking triggers existential presupposition. For example, assuming that focus-marking triggers an existential presupposition, (40a) has the meaning in (41) – i.e., given that John teaches something on Thursday, the most desirable worlds are where he teaches logic; so, given the context, we predict that (40a) is false.

$$(41) \quad \llbracket (40a) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \exists x[\text{John teaches } x \text{ on Thursdays}] \\ 1 \text{ iff defined and } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \\ \quad \llbracket \text{John teaches logic on Thursdays} \rrbracket(w') = 1 \end{cases}$$

In contrast, we predict that (40b) is true because it has the meaning in (42) – i.e., given that John teaches logic on some day, the most desirable worlds are where he teaches logic on Thursday.

$$(42) \quad \llbracket (40b) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \exists x[\text{John teaches logic on } x] \\ 1 \text{ iff defined and } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \\ \quad \llbracket \text{John teaches logic on Thursdays} \rrbracket(w') = 1 \end{cases}$$

Hence, my indirect restriction semantics predicts that the focus-sensitivity of priority modals is simply a byproduct of how they interact with local contexts and presuppositions.

Moreover, as I mentioned in §1, the focus-sensitivity of *want* can help us explain various counterexamples to UM which involve *want* taking complements with adverbial modifiers – e.g., Stalnaker’s example in (10) involves a prejacet modified by the adverb *peacefully*. We can blame these counterexamples to UM on presupposition too. It’s been noted in the literature that adverbial modifiers like *late* and *fast* trigger inferences which exhibit the same projective behavior as presuppositions (Simons, 2001; Schlenker, 2008; Abrusán, 2013).

- (43) a. John didn’t arrive late. $\rightsquigarrow$ *John arrived* [negation]
 b. Did John arrive late? $\rightsquigarrow$ *John arrived* [question]
 c. John probably arrived late. $\rightsquigarrow$ *John arrived* [modal]
- (44) a. Alice didn’t run fast. $\rightsquigarrow$ *Alice ran* [negation]
 b. Did Alice run fast? $\rightsquigarrow$ *Alice ran* [question]
 c. Alice probably ran fast. $\rightsquigarrow$ *Alice ran* [modal]

Abrusán (2013) argues that this presupposition arises because the relevant modifier is focus-marked. This hypothesis seems plausible for a few reasons. First, while English and many other languages mark focus via heavy stress or a pitch accent on the focused constituent, languages like Hungarian mark focus via special word order (Szabolcsi, 1981; Szendroi, 2001). Abrusán (2013) shows that, in Hungarian, modifiers only trigger a presupposition when the syntax signals that the modifier is focused. Second, as Simons (2001) observes, even in languages like English the presupposition of adverbial modifiers can be canceled by placing focal stress on some other constituent. If we assume that *peacefully* in Stalnaker’s example is focus-marked, my semantics predicts that (10) has the following truth-conditions:

$$(45) \quad \llbracket (10) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \exists x[\text{Mary dies } xly] \\ 1 \text{ iff defined and } \forall w' \in \text{BEST}(w, f^{+\kappa}, g) : \llbracket \text{Mary dies peacefully} \rrbracket(w') = 1 \end{cases}$$

Hence, we predict the desired non-entailment, because the sentence “Mary wants to die peacefully” quantifies only over the worlds in the modal base where Mary dies in some way, whereas “Mary wants to die” quantifies over the entire modal base.

Our semantics for deontic modals, when coupled with the assumption that adverbial modifiers are focus-marked, also yields an intuitive solution to the Samaritan Paradox’s adverbial cousin – the Paradox of Gentle Murder (Forrester, 1984). Consider the following argument:

- i. Smith ought not murder Jones. [premise]
- ii. If Smith murders Jones, Smith ought to murder Jones gently. [premise]
- iii. Smith murders Jones. [premise]
- iv. Smith ought to murder Jones gently. [MP: ii, iii.]
- v. Smith ought to murder Jones. [UM: iv.]

In nearly all contexts, the premises (i.) and (ii.) are both true and non-contradictory; hence, if, contrary to the law, Smith does murder Jones, then (iv.) follows from (ii.) and (iii.) by *modus ponens*.²¹ Given UM, we can then derive (v.) from (iv.); however, this implies a contradiction as it is inconsistent with (i.) given the following minimal assumptions:²²

- i. **Agglomeration.** $\bigcirc\varphi, \bigcirc\psi \models \bigcirc(\varphi \wedge \psi)$
- ii. **D Axiom.** $\bigcirc\varphi \models \Diamond\varphi$

²¹The validity of this step depends on substantive questions regarding both the semantics and syntax of conditionals. First, given a variably-strict semantics for the conditional, *modus ponens* is only valid assuming the selection function satisfies Weak Centering. Second, standard implementations of Kratzer’s (1986, 2012) restrictor theory of conditionals’ syntax-semantic interface invalidate *modus ponens*. I return to both of these issues in the next section.

²²While the modal D Axiom is fairly uncontroversial, Jackson (1985) presented several famous counterexamples to AGGLOMERATION; however, a number of authors have raised serious concerns about Jackson’s argument (Cariani, 2016; Boylan, 2023) and I will take AGGLOMERATION for granted in this paper.

However, given the perspective adopted here, the argument from (i.)-(iii.) to (v.) does not go through because the key instance of UM is invalid. If we assume that *gently* bears focus-marking, then *ought* in (iv.) only quantifies over a modal base that satisfies the existential presupposition triggered by focus – i.e., worlds where there exists some way in which Jones dies. In contrast, *ought* in (v.) quantifies over all the worlds in the modal base because its prejacent is presupposition-free.

Application #3: restriction in conditionals and Chisholm’s Paradox Besides its nice prediction concerning disjunction, my indirect restriction semantics also makes interesting predictions concerning the interaction of priority modals and conditionals. Since a thoroughgoing discussion of deontic conditionals is outside the current paper’s scope, I will focus my attention on one famous problem: Chisholm’s Paradox. Chisholm (1963) highlighted a tension between two seemingly natural principles governing conditional *oughts*:²³

(46) **Factual Detachment (FD).**

$$\text{if } \varphi, \bigcirc\psi, \varphi \models \bigcirc\psi$$

(47) **Deontic Detachment (DD).**

$$\text{if } \varphi, \bigcirc\psi, \bigcirc\varphi \models \bigcirc\psi$$

Here’s a slightly updated version of Chisholm’s (1963) famous vignette:

Jones’s neighbors need help with lawn care. John has a riding mower and he can mow his neighbors’ lawn minimally inconveniencing himself. It seems that Jones ought to go and help his neighbors with their lawn and Jones ought to tell them he’s coming if he goes over to help them with their lawn. However, if he doesn’t go over to help his neighbors, Jones should not tell his neighbors that he’s coming. Finally, let’s suppose Jones, in fact, does not go to help his neighbors.

Let’s formalize Chisholm’s vignette as the following four sentences:

²³My presentation differs slightly from how Chisholm’s Paradox is often presented in the literature. FACTUAL DETACHMENT is often formulated in terms of $\bigcirc$ taking wide-scope over the conditional, and the conditional in (48b) is typically translated as a deontic modal taking wide-scope over the conditional. Saint Croix and Thomason (2014), however, provide what I believe to be conclusive arguments that deontic modals taking wide-scope over conditionals are largely unattested in natural language. See Cariani (2021) for a useful overview of these issues.

- (48) a. $\bigcirc go$
 b. $if\ go, \bigcirc tell$
 c. $if\ \neg go, \bigcirc \neg tell$
 d. $\neg go$

These four claims are all intuitively true. However, a very simple argument shows that, given both FACTUAL DETACHMENT and DEONTIC DETACHMENT, these four claims are jointly inconsistent. First, we can $\bigcirc tell$ from (48a) and (48b) via DD and we can derive $\bigcirc \neg tell$ from (48c) and (48d) via FD. If we assume AGGLOMERATION and the modal D Axiom, then $\bigcirc tell$ and $\bigcirc \neg tell$ together entail a contradiction. Hence, we seemingly have to abandon one of DD or FD.²⁴

Kratzer's (1986, 2012) restrictor analysis of *if*-clauses invalidates FACTUAL DETACHMENT while preserving DEONTIC DETACHMENT. Kratzer – building on earlier ideas in Lewis (1975) and Heim (1982) – argues that *if*-clauses are devices for restricting the domain of quantification of operators. Kratzer gave the following provocative description of her view:

The history of the conditional is the story of a syntactic mistake. There is no two-place *if...then* connective in the logical forms of natural language. *If*-clauses are devices for restricting the domains of various operators. (p. 11)

Lewis's original idea is that conditionals function analogously to binary nominal quantifiers which combine first with their restrictor and then with a matrix predicate. Similarly, Lewis thought that an *if*-clause first restricts the domain of an operator, such as adverbial quantifier, and then combines with the matrix proposition. Hence, according to the restrictor analysis, a deontic conditional has an LF like the one in FIGURE 1. In contrast to Lewis, Kratzer proposed a semantic implementation of the restrictor analysis where the *if*-clause restricts the modal base:

$$(49) \quad \llbracket if \rrbracket^{w,f,g} = \lambda p_{\langle s,t \rangle} \lambda q_{\langle s,t \rangle} . \forall w' \in f^{+p}(w) : q(w') = 1$$

When combined with the naïve Kratzerian semantics for priority modals, the restrictor analysis predicts that (48a)-(48d) are consistent by invalidating FACTUAL DETACHMENT. For example, in

²⁴A version of Chisholm's paradox also crops up with desire ascriptions – see Lauer and Condoravdi (2014) for discussion of these cases.

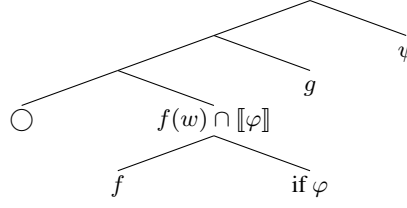


Figure 1: LF for deontic conditional according to the restrictor view

Chisholm’s vignette, the following ordering source seems natural:

$$w_{GT} \succ w_{G\bar{T}} \succ w_{\bar{G}\bar{T}} \succ w_{\bar{G}T}$$

Given this ordering source, when restricted to $\neg go$ -worlds the modal base is the set $\{w_{\bar{G}\bar{T}}, w_{\bar{G}T}\}$ and the modal domain is the singleton $\{w_{\bar{G}\bar{T}}\}$; so, the conditional in (48b) is true, but $\bigcirc \neg tell$ is false because it’s evaluated against the unrestricted modal domain $\{w_{GT}\}$. Hence, the restrictor analysis, when coupled with the naïve Kratzerian semantics, predicts that FACTUAL DETACHMENT fails (similar reasoning shows that DEONTIC DETACHMENT is valid).

As previously noted, my indirect restriction semantics for priority modals mirrors the restrictor analysis closely – the process by which the domain of priority modals are restricted to a local context satisfying their prejacent’s presuppositions is exactly the same as the process by which *if*-clauses restrict the domain of a modal in their consequent. This has a nice upshot: we can recover a Kratzer-style semantics for prioritized conditionals from just our assumptions about how priority modals interact with local contexts. Hence, even if you do not view treating conditionals as two-place connectives as a “syntactic mistake,” we can still replicate the logic of the restrictor analysis in our semantics, including the predicted failure of FACTUAL DETACHMENT.

For example, let’s assume the material analysis of conditional. Given the material analysis, Schlenker’s algorithm for local contexts predicts that the consequent’s local context is the smallest proposition κ such that the following holds for all well-formed completions ψ' :

$$\varphi \supset (\kappa \wedge \psi') \Leftrightarrow \varphi \supset \psi'$$

It's straightforward to show that, given the material analysis, Schlenker's algorithm predicts the consequent's local context κ is simply the φ -worlds in c . Likewise, if we instead assume a strict conditional analysis of the conditional, then the consequent's local context is the smallest proposition κ such that the following holds for all well-formed completions ψ' :

$$\Box(\varphi \supset (\kappa \wedge \psi')) \Leftrightarrow \Box(\varphi \supset \psi')$$

It's straightforward to show that the consequent's local context is $\kappa = \bigcup \{f^{+\varphi}(w) : w \in c\}$ – i.e., the φ -worlds in the modal base accessible from the worlds in the context set. In both cases, my indirect restriction semantics replicates the prediction of restrictor analysis coupled with the naïve Kratzerian semantics for priority modals: FACTUAL DETACHMENT is invalid. If we assume a variably-strict semantics, like the one in (50), then computing the consequent's local context via Schlenker's algorithm becomes slightly more complicated.

$$(50) \quad \llbracket \varphi > \psi \rrbracket^{w,f,g} = 1 \Leftrightarrow \forall w' \in \text{MIN}(w, f^{+\varphi}, g) : \llbracket \varphi \rrbracket^{w,f,g} = 1$$

Mackay (2019) shows that, if we drop the Limit Assumption, then the local context of a variably-strict conditional's consequent is not defined. However, given the Limit Assumption, it's easy to show that the local context of a variably strict conditional's consequent is $\kappa = \bigcup \{\text{MIN}(w, f^{+\phi}, g) : w \in c\}$. Hence, assuming a variably-strict conditional, my indirect restriction semantics also predicts that FACTUAL DETACHMENT fails.

Application #4: the absence of an epistemic Samaritan Paradox In contrast to priority modals, Booth (2022) observes that epistemic modals appear to always license Samaritan-like inferences as examples like (51) demonstrate.

- (51) a. John might have helped the person who was murdered.
 $\Rightarrow$ There might have been a mugging.
 b. Susie must regret punching Timmie.
 $\Rightarrow$ Susie must have punched Timmie.

This difference, I believe, stems from differences in how local contexts interface with different

flavors of modality. Whereas local contexts *restrict* the domain of priority modals, local contexts *bound* the domain of epistemic modals. To explain Yalcin’s (2007) observation that “ $\neg\varphi$ and might φ ” and similar constructions are semantic contradictions, Mandelkern (2019, 2024) argues that local contexts bound the admissible domains of quantification for epistemic modals.²⁵

$$(52) \quad \llbracket \text{might} \rrbracket^{w,f,g,k} = \lambda p_{\langle s,t \rangle} \cdot \begin{cases} \text{defined if } f(w) \subseteq \kappa \\ 1 \text{ iff defined and } \exists w' \in f(w) : p(w') = 1 \end{cases}$$

If we combine Mandelkern’s bounded semantics for epistemics with the analysis of factive attitude verbs from the previous section, (51b) has the following truth-conditions:

$$(53) \quad \llbracket (51b) \rrbracket^{w,f,g,\kappa} = \begin{cases} \text{defined if } \kappa \subseteq \mathbf{punched}(s, t) \wedge f(w) \subseteq \kappa \\ 1 \text{ iff defined and } \forall w' \in f(w) : \mathbf{regret}(\mathbf{punching}(s, t)) = 1 \end{cases}$$

These truth-conditions in (53) entail that Susie must have punched Timmie, because epistemic *must*’s bound presupposition forces the modal base to only contain worlds where Susie punched Timmie. As a result, Mandelkern’s bound presupposition ensures that epistemic modals always validate Samaritan-like inferences consonant with the data.

Moreover, if local contexts restrict the domain of priority modals, while they bound the domains of epistemic modals, this difference provides us with a useful diagnostic for determining a modal’s flavor. For example, in English, the weak necessity modals *ought* and *should* have a “quasi-epistemic” use, exemplified by (54).

(54) It’s three. John ought to be in his office.

In such uses, “ought φ ” is intuitively true just in case φ is expected, or would occur in the normal course of events. However, it’s controversial whether this reading is truly epistemic. Copley (2003, 2006) and Yalcin (2016) have both argued that the quasi-epistemic uses of *ought* and *should* are not truly epistemic. The fact that epistemic modals validate Samaritan-like inferences provides us with insight into this debate. Unlike paradigmatic epistemic modals, the quasi-epistemic use of *ought*

²⁵Mandelkern’s (2019, 2024) account of Yalcin’s data crucially assumes that local contexts are symmetric, whereas we’ve assumed that local contexts are asymmetric.

doesn't appear to license the Samaritan inference.²⁶

- (55) a. John ought to have survived the fall.
 $\nRightarrow$ John ought to have fallen.
 b. John must have survived the fall.
 $\Rightarrow$ John must have fallen.

The inference in (55a) is not valid given a quasi-epistemic reading of *ought*, because John's survival may have been expected even if the fall itself was unexpected. This supports Copley and Yalcin's thesis that the quasi-epistemic reading of *ought* is not a genuine epistemic modal.

4.1 The Local Zvolenszky Problem

When paired with a naïve Kratzerian semantics for priority modals, the restrictor analysis of conditionals generates a well-known bug: it renders "if φ , ought φ " trivially true (Zvolenszky, 2002; Carr, 2014; Chung, 2020). However, unlike, say, epistemic conditionals of the form "if φ , must have φ ," these conditionals do not seem trivially true:

- (56) a. If Britney Spears drinks Pepsi, she ought to drink Pepsi. [deontic]
 b. If Britney Spears drank Pepsi, she must have drunk Pepsi. [epistemic]

Given my analysis of the Samaritan Paradox, an analogous problem arises in my system, what I'll call the *local Zvolenszky problem*. For example, in (57), the local context of the right conjunct consists of the worlds in the context set where someone was murdered; hence, my theory predicts that the second conjunct of (57) should be trivially true.

- (57) John was murdered and someone ought to be murdered.

In addition to predicting that "if φ , ought φ " is trivially true, Carr (2014) observes that Kratzer's semantics also wrongly predicts that "if φ , ought not φ " is trivially false. She illustrates this with the "Death in Damascus" scenario from Gibbard and Harper (1978):

If you are in the same city as Death tomorrow, you'll die. Death has planned to be

²⁶Thanks to Fabrizio Cariani (p.c.) for this observation.

wherever he predicts you'll be, and he's very reliable in such predictions. Your options are to stay in Damascus or go to Aleppo. But, as you know, if you stay in Damascus, then that's good evidence that Death will be there. Similarly for going to Aleppo.

In this context, the following conditionals both seem true:

- (58) a. If you're in Damascus, you shouldn't be in Damascus.
 b. If you're in Aleppo, you shouldn't be in Aleppo.

A similar problem arises given my account. For example, in (59), the modal base is restricted to a set of worlds where John didn't wash his hands; so, the indirect restriction account predicts that (59) is trivially false.

- (59) John didn't wash his hands, but he ought to have washed his hands.

These may seem like serious defects of the current theory. However, the local Zvolenszky problem is not only a problem for my theory. It's also a problem for nearly all dynamic analyses of deontic modality. Since my analysis is built on top of a static translation of dynamic semantics, this is unsurprising. In Willer's (2014, 2016) dynamic semantics, a *deontic context* d determines for every context $c \in \mathcal{C}$ a deontically ideal state c_d by providing an ordering source $o \subseteq \mathcal{C}^{\mathcal{C}}$. We can then give deontic necessity modals a Veltman-style (1996) test semantics:

$$c[\Box_d \varphi] := \{w \in c : c_d[\varphi] = c_d\}$$

However, if we adopt a standard dynamic entry for conjunction, then Willer's semantics will generate instances of the local Zvolenszky problem. Willer blocks this prediction by adopting a version of DIVERSITY, imposing the requirement that input contexts are “non-trivial” in the sense that its prejacent as well as its negation must be open possibilities” (p. 345).

DIVERSITY is commonly invoked to solve the Zvolenszky problem. Frank (1997), for example, proposed that “if φ , ought φ ” is infelicitous because, when restricted to φ -worlds, the domain of *ought* fails to satisfy DIVERSITY. However, Zvolenszky (2002) observed that DIVERSITY-based solution faces what she called the “flipside problem,” because they predict that “if φ , ought φ ” is

always defective. However, this prediction is incorrect. Despite being instances of “if φ , ought φ ,” the following conditionals are felicitous:

- (60) a. If the Dalai Lama is mad, then he should be mad. (Zvolenszky, 2002)
 b. If John turns on Exit 49, then he should turn on Exit 49. (Carr, 2014)

These conditionals are coherent and seemingly true: (60a) is true because, if the Dalai Lama is mad about something, then there’s a good reason for him to be mad about the thing in question, and (60b) is true because, if John takes Exit 49, then there’s a good reason for him to take Exit 49. Given that DIVERSITY predicts that these conditionals will always be unassertible, DIVERSITY isn’t the right way to solve Zvolenszky’s problem.

In sum, while the local Zvolenszky problem is a problem, it’s simply an instance of a problem that any theory of deontic modality must solve anyway. A number of extant solutions to Zvolenszky’s problem can be integrated with my solution to the Samaritan Paradox. For example, in other work, I have proposed a solution to the “if φ , ought φ ” problem that involves enriching the ordering source with default rules from non-monotonic logic (REDACTED, xxxx).²⁷ However, the local Zvolenszky problem forecloses the possibility of giving a unified explanation of the Samaritan Paradox and the “if φ , ought φ ” problem.²⁸ If I’m correct, then these two problems require separate modular solutions rather than a single unified solution.

5 Conclusion

It has often been observed that the Samaritan Paradox and related puzzles are intimately connected to presupposition. In this paper, I have taken that observation seriously, arguing that priority modals are interpreted relative to modal domains that are locally restricted by the presuppositions of their prejacent. Once this interaction between presupposition and modal domain restriction is recognized, the Samaritan Paradox and related puzzles no longer provide evidence for non-monotonicity.

²⁷ A similar reason-based analysis has also been defended by Chung (2020) employing causal models rather than default rules. However, Chung’s solution relies on Geurts’s (2004) strategy of allowing the *if*-clause in a deontic conditional to restrict a cover epistemic necessity modal rather than the overt deontic modal – so, it’s unclear if his solution can be generalized to cover cases of the Zvolenszky problem when it arises in conjunction and disjunction.

²⁸ Thanks to Una Stojnić (p.c.) for highlighting this objection.

The proposal also yields immediate solutions to a range of deontic paradoxes, including the Paradox of Gentle Murder and Chisholm's Paradox, while generating several further desirable predictions. In particular, it predicts both the focus sensitivity of priority modals and the absence of an epistemic Samaritan Paradox.

At the same time, the view comes with a significant cost: it leads to a proliferation of the Zvolenszky problem. If the present account is correct, then the Zvolenszky problem is not merely a problem about the interaction between conditionals and deontic modals. Rather, it reflects a more general cross-categorical challenge arising from the interaction of presupposition-sensitive domain restriction with logical connectives more broadly. I leave this problem for future work.

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